Recommendations: Churn Prediction Proof of Viability

STADIOchoice Streaming Churn Prediction, CAP182 Capstone SS2, Part D

This report interprets the results presented in Part C, where a Logistic Regression model and an XGBoost model were trained and compared on the Telco Customer Churn dataset as a proof of viability ahead of receiving STADIOchoice's own subscriber data. It addresses which model should be carried forward, how that model can be strengthened, what will need to change when it meets the real data requested in SS1, and whether these findings agree with the literature reviewed in Part A.

1. Which model is best between Model 1 and Model 2?

XGBoost is the better model and is recommended as the one to carry forward. It outperformed Logistic Regression on every metric measured: accuracy (0.753 versus 0.734), precision (0.523 versus 0.499), recall (0.802 versus 0.794), F1 score (0.633 versus 0.613), and ROC-AUC (0.844 versus 0.842). More importantly, this was not a marginal, coincidental edge. An exact McNemar's test on the paired predictions from the same test set returned a p-value of 0.0108, meaning the difference in the two models' errors is statistically significant at the 0.05 level. XGBoost corrected 66 customers that Logistic Regression got wrong, against only 39 the other way, a meaningfully lopsided split. For a retention use case, the practical gain matters as much as the statistical one: higher precision at a similar recall means fewer retention offers wasted on customers who were never going to leave, which is precisely the inefficiency identified in the original STADIOchoice problem statement.

2. How can the selected model be improved?

The current XGBoost model used sensible default-adjacent hyperparameters rather than a tuned configuration, so a systematic search over tree depth, learning rate, and the number of estimators, using cross-validation rather than a single train/test split, is the most direct next step and would likely close some of the gap to the strongest results in the literature. Beyond tuning, three further improvements are worth pursuing. First, probability threshold calibration: STADIOchoice's own problem statement asks for a ranked, actionable list of at-risk households rather than a binary flag, so the model's output should be treated as a risk score and the cutoff for action tuned to the retention team's actual capacity and cost tolerance, rather than left at the default 0.5. Second, class imbalance handling deserves more experimentation. This project used class weighting for simplicity, but Sikri et al. (2024) found that the choice and ratio of balancing strategy materially changes results, so testing SMOTE oversampling and ratio-based balancing against the current approach is a reasonable next step. Third, adding SHAP-based explainability, as El Attar and El-Hajj (2026) did, would let the model's flagged households come with a reason attached, which matters for a retention team that needs to act on, not just trust, the model's output.

3. How will the selected model need to be adapted to work with your requested data from SS1?

The Telco dataset proved the approach, but STADIOchoice's actual data, as scoped in the SS1 data request, is both richer and structured differently, and the model will need real adaptation rather than a straight swap of input files. The clearest gain is on the behavioural side: Telco offers only static service flags (for example, whether a customer has streaming TV), while STADIOchoice's streaming viewing logs provide timestamped play, pause, search, and abandonment events per household. These support far stronger engineered features than anything available here, such as recent viewing trend, genre diversity, and exposure to sports content, which STADIOchoice's own briefing pack identifies as a specific churn trigger around tournament endings, a signal this proof of concept cannot demonstrate because the public dataset has no equivalent. The target variable will also need rethinking: Telco's churn label is a single binary flag, but STADIOchoice's cancellation and retention data separately captures the cancellation request, the reason, and the retention offer outcome, and the original problem statement specifically flags that customers who quietly stop watching without ever cancelling are currently missed entirely. Capturing that group will likely require a behavioural definition of churn in addition to, not instead of, the contractual cancellation date. Customer support free-text notes are new territory altogether, since the Telco dataset contains no text data, so using that source at all will require adding a text preprocessing step this project did not build. Finally, the class imbalance in the real data is different: STADIOstream's actual monthly churn rate is around 9.2%, compared to 26.5% in the Telco dataset, so the scale_pos_weight and class weighting used here cannot simply be reused and will need to be recalculated once real data is available, alongside a full retraining and re-tuning of both models rather than treating the Telco-derived hyperparameters as final.

4. Does your results align with the related literature from Part A?

Broadly, yes, on the qualitative finding, with an honest gap on the exact numbers. All three papers reviewed in Part A found ensemble or boosting methods outperforming simpler linear models on churn prediction, and this project reproduces that finding directly, with a statistically significant XGBoost advantage over Logistic Regression. On ROC-AUC specifically, this project's results (0.842 to 0.844) sit close to Lalwani et al.'s (2022) benchmark of 84% AUC on the identical dataset. On accuracy, this project's XGBoost result (0.753) falls short of both Lalwani et al.'s AdaBoost result (81.71%) and El Attar and El-Hajj's (2026) XGBoost result (0.84), and that gap has a defensible explanation rather than being unexplained: this project deliberately used a simpler, more directly comparable feature set and class weighting instead of the SMOTE oversampling and considerably more extensive feature engineering (46 features, against 36 here) that El Attar and El-Hajj applied, and did not perform hyperparameter tuning. That gap is itself informative, since it points at the two most valuable next steps identified in Question 2: tuning and more deliberate imbalance handling.

References

El Attar, A. and El-Hajj, M. (2026) 'Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis', Frontiers in Artificial Intelligence, 9, 1748799.

Lalwani, P., Mishra, M.K., Chadha, J.S. and Sethi, P. (2022) 'Customer churn prediction system: a machine learning approach', Computing, 104(2), pp.271-294.

Sikri, A., Jameel, R., Idrees, S.M. and Kaur, H. (2024) 'Enhancing customer retention in telecom industry with machine learning driven churn prediction', Scientific Reports, 14, 13097.